DSML EXAM PREPARATION GUIDE

Module-wise important topics • 4-mark and 8-mark priorities
Updated using the Series Test I paper and uploaded previous-year question papers

How to use this guide

This guide combines the topics in your uploaded DSML module notes, the Series Test I paper, and the DSML questions visible in *S3 previous year question papers.pdf*. Priority is based on topic recurrence and the kinds of questions asked; it is not a guarantee of the next question paper.

ANSWER TYPE	YOUR REQUIRED LENGTH	HOW TO PREPARE
4 marks	About 2 handwritten pages	Definition, core explanation, key points/steps, small example or diagram, short conclusion.
8 marks	About 3–4 handwritten pages	Definition, detailed explanation, labelled diagram/flowchart, steps or formula, worked example where relevant, advantages/applications.

Important: The previous-year DSML papers use 4 marks for Part A and 8 marks for Part B. The Series Test I paper uses 3-mark and 6-mark questions, so the study priorities below distinguish previous-year evidence from the series-test evidence.

Priority key

VERY HIGH	Repeated in previous papers and/or directly tested in Series Test I; prepare a full answer and practice the numerical.
HIGH	Strong long-answer or short-answer candidate; prepare definitions, steps, formulas and examples.
MEDIUM	Prepare a compact answer after finishing very-high and high priority topics.

What the previous-year papers tell us

The DSML papers are visible on pages 1–7 of the uploaded 13-page PDF. The remaining pages belong to other courses, so they were not used for DSML topic ranking.

PAPER PAGE	QUESTION THEMES VISIBLE	STUDY IMPLICATION
1–3	Descriptive vs inferential statistics; visualization; Data Science process; KNN; Naive Bayes with Laplace smoothing; decision-tree pruning; linear regression; SVM; neural networks; confusion matrix; ensemble learning.	Prioritize Data Science process, visualization, KNN, Naive Bayes, pruning, regression, SVM and evaluation metrics.
4–7	Feature selection; classification rules from decision trees; backpropagation; K-means; non-spherical clusters; multivariate plots; KNN; Naive Bayes with Laplace smoothing; entropy and best attribute selection; divide-and-conquer vs separate-and-conquer; SVM; confusion matrix.	KNN and Naive Bayes recur; add entropy/information gain, clustering, feature selection, decision-tree rules and neural-network topics to revision.

Series Test I cross-check

The Series Test I paper directly includes mean/variance/standard deviation, classification vs regression, reinforcement learning, data storage/abstraction/generalization/evaluation, the Data Science process, visualization, KNN (k = 3) and Naive Bayes. This supports giving these topics extra attention.

Topics seen in the previous-year papers but not clearly covered in the supplied Module 1–5 PDFs

The previous-year paper also asks about ordinary least squares/linear regression, feature selection, neural-network topologies and backpropagation, bagging/boosting/Random Forests, and confusion-matrix metrics. These are included below because you asked to consider the past papers. Cross-check your official syllabus or class notes before treating them as part of a specific module.

Module 1 — Data Science Fundamentals

Focus: statistics, the Data Science process, data visualization, and univariate/multivariate analysis.

TOPIC	PRIORITY	4-MARK PREP	8-MARK PREP
Data Science process + diagram	VERY HIGH	List stages and purpose briefly.	Explain each stage with a flowchart and one running example.
Data visualization techniques	VERY HIGH	Define visualization and explain selected charts.	Explain visualization methods; include examples and uses. Cover univariate and multivariate plots.
Univariate vs multivariate analysis	VERY HIGH	Definitions, differences and examples.	Explain techniques, relationships, plots and interpretation.
Mean, median, mode, range, variance, SD	HIGH	Definitions and formulas.	Solve a full numerical if asked; show deviations, squares and final answer.
Descriptive vs inferential statistics	HIGH	Compare with examples.	Expand into purpose, methods and examples if asked.
Classification vs regression	HIGH	Definition, output type, examples and differences.	Add workflow, use cases and comparison table.
Dataset and role in DS/ML	MEDIUM	Definition and role.	Use if combined with data collection, preparation and modelling.
OLS / linear regression	PYQ-BASED	Equation and coefficient meaning.	Prepare interpretation of coefficients and prediction example; verify in class notes.

Numerical practice

Practise mean, variance and standard deviation using the exact style of the Series Test I question. For 8-mark preparation, show each calculation step clearly rather than writing only the final value.

Past-paper signal

Previous-year DSML pages 1 and 5 ask about visualization and the Data Science process; the Series Test I paper also asks the Data Science process and visualization as long-answer alternatives.

Module 2 — Machine Learning Algorithms

Focus: how machines learn, learning types, KNN, probabilistic learning, Naive Bayes and Laplace smoothing.

TOPIC	PRIORITY	4-MARK PREP	8-MARK PREP
K-Nearest Neighbour (KNN)	VERY HIGH	Definition, algorithm steps and Euclidean distance.	Full numerical: calculate distances, sort neighbours, select k and vote for class.
Naive Bayes + Laplace smoothing	VERY HIGH	Bayes' theorem, priors, likelihood and smoothing idea.	Full PlayTennis-style example: priors, smoothed conditional probabilities, joint/posterior values and final class.
Types of Machine Learning	HIGH	Compare supervised, unsupervised, semi-supervised and reinforcement learning.	Explain all four with workflow, algorithms and examples.
Data storage, abstraction, generalization, evaluation	HIGH	Explain all four stages with a simple example.	Use a process diagram and describe how each stage contributes to learning.
Probabilistic learning / Bayes theorem	HIGH	Define probabilities and state the theorem.	Show the calculation sequence within a Naive Bayes solution.
Laplace estimator	HIGH	State why smoothing is used and its formula from notes.	Apply smoothing to zero-count categorical features in a worked example.

Numerical practice checklist

• KNN: compute distance from the unknown point to every training point, order the distances, select k = 3 when specified, and use majority voting.

• Naive Bayes: calculate prior probability for each class; calculate each conditional probability; multiply the terms; compare class scores.

• Laplace smoothing: apply it to the categorical probability counts when the question explicitly requests smoothing.

Past-paper signal

KNN appears as a long numerical question in both the Series Test I paper and previous-year DSML pages 1 and 5. Naive Bayes with Laplace smoothing appears in previous-year pages 2 and 5, and in the Series Test I paper.

Module 3 — Decision Tree Learning

Focus: tree construction, entropy, information gain, Gini, pruning and classification rules.

TOPIC	PRIORITY	4-MARK PREP	8-MARK PREP
Entropy + Information Gain	VERY HIGH	Define impurity and state formulas.	Calculate dataset entropy, weighted entropy and information gain for candidate attributes; select the best split.
Decision Tree construction	VERY HIGH	Root, decision nodes, branches and leaves.	Build a tree from a dataset and explain each split.
Gini Index	HIGH	Definition and formula.	Calculate Gini for each candidate split and compare weighted impurity.
Pruning: pre- and post-pruning	VERY HIGH	Definitions, differences and purpose.	Explain both methods, steps, advantages and overfitting prevention.
Classification rules / Separate-and-Conquer	HIGH	IF–THEN rule, antecedent and consequent.	Derive rules from data/tree; compare separate-and-conquer with divide-and-conquer.
ID3, C4.5, C5.0; Gain Ratio	HIGH	Compare splitting criteria and capabilities.	Explain algorithms, attribute selection and pruning differences.

Formula reminders

Entropy	$H(S) = -\Sigma\ p_{\blacksquare}\ \log_{\blacksquare}(p_{\blacksquare})$
Information Gain	IG(S,A) = Entropy(S) – Weighted Entropy(S,A)
Gini Index	$Gini(S) = 1 - \Sigma\ p_{\blacksquare}^2$

Past-paper signal

Previous-year DSML pages 2, 5 and 6 include pruning, entropy/attribute selection and classification rules. These are strong 8-mark preparations; do not study only definitions.

Module 4 — SVM and PYQ-linked model topics

The supplied Module 4 PDF is focused on Support Vector Machines. The previous-year DSML papers also contain questions on linear regression, neural networks, backpropagation, feature selection and ensemble methods. These extra topics are marked PYQ-based because their exact module placement is not established by the visible module PDFs.

TOPIC	PRIORITY	4-MARK PREP	8-MARK PREP
SVM working + diagram	VERY HIGH	Definition, classification use and decision boundary.	Explain training points, separating hyperplane, margin and support vectors with a labelled diagram.
Maximum-margin hyperplane	VERY HIGH	Define margin and support vectors.	Explain the maximum-margin idea, three parallel lines and why closest points define the boundary.
Hyperplane equation and classification	HIGH	w■x + b = 0; explain weights and bias.	Substitute sample values and identify which side/class the point belongs to.
Linear regression / OLS	PYQ-BASED	Equation and coefficient interpretation.	Explain a fitted regression equation, interpret coefficients and calculate a prediction.
Neural-network topologies	PYQ-BASED	Single-layer feedforward, multilayer feedforward, recurrent.	Describe each topology with a diagram and application.
Backpropagation	PYQ-BASED	Purpose and role of learning rate.	Explain forward pass, error, backward gradient flow and weight update; follow class notes.
Bagging, Boosting, Random Forests	PYQ-BASED	Definitions and key differences.	Compare working, overfitting, suitability and applications.
Feature selection	PYQ-BASED	Purpose and why it matters.	Explain why it is necessary before model building; include methods only if in syllabus notes.

Diagram checklist for SVM

Draw two classes on a feature plane; add the central decision hyperplane and two parallel margin boundaries; circle the nearest data points as support vectors. Label the equation w■x + b = 0 and explain that the weight vector controls orientation while b shifts the boundary.

Past-paper signal

Previous-year DSML pages 2–4 and 6–7 ask about the SVM margin, support vectors and hyperplane, and also ask about neural networks, regression, feature selection and ensemble learning.

Module 5 — Clustering

Focus: K-means, centroid updates, distance measures, elbow method and hierarchical clustering.

TOPIC	PRIORITY	4-MARK PREP	8-MARK PREP
K-means algorithm	VERY HIGH	Definition and six steps.	Complete worked example: choose K, initialize centroids, calculate distances, assign clusters, update centroids and repeat.
Euclidean distance + centroid	VERY HIGH	State formula and terms.	Show distance calculations and centroid mean calculations for a dataset.
Elbow method	HIGH	Purpose, WCSS and how to find elbow.	Explain procedure and interpret a small K-vs-WCSS table/plot.
Hierarchical clustering	HIGH	Define dendrogram and agglomerative/divisive methods.	Explain both approaches with steps, dendrogram and comparison.
K-means limitations / non-spherical clusters	PYQ-BASED	Why K-means may fail for non-spherical clusters.	Discuss limitations and suggest alternatives only if covered in your syllabus/class notes.
Clustering types and applications	MEDIUM	Definitions and examples.	Compare hierarchical and partitioning methods; include use cases.

Numerical practice checklist

- Calculate Euclidean distance from every point to each centroid.
- Assign each point to the nearest centroid.
- Calculate new centroid coordinates using the mean of cluster coordinates.
- Repeat until assignments/centroids stop changing; state the final clusters.

Past-paper signal

Previous-year DSML pages 5–7 ask about K-means, the effect of distance measure and number of clusters, and why K-means can fail on non-spherical clusters.

Final revision plan

Complete the high-priority topics first, then use the last revision session for formula recall and timed practice.

STAGE	WHAT TO DO	TARGET OUTPUT
1. Core long answers	Data Science process; visualization; types of ML; KNN; Naive Bayes; entropy/IG; pruning; SVM; K-means.	Prepare one full 8-mark answer for each.
2. Numericals	Mean/variance/SD; KNN; Naive Bayes + Laplace; entropy/IG; Gini; K-means; confusion matrix and metrics.	Solve each without looking at the notes.
3. Short answers	Definitions, comparisons, algorithms, advantages/limitations and formulas across all modules.	Prepare 4-mark answers to the required length.
4. PYQ check	Attempt the DSML questions visible on pages 1–7 of the uploaded previous-year PDF.	Mark each answer: confident / revise / not prepared.

Most important numerical topics

TOPIC	MUST SHOW IN THE ANSWER
KNN	Distance table → sorted neighbours → chosen k → majority vote → predicted class.
Naive Bayes	Class priors → conditional probabilities → Laplace smoothing if requested → class scores → final prediction.
Entropy / Information Gain	Class counts → initial entropy → subset entropies → weighted entropy → information gain → best attribute.
Gini Index	Class proportions → Gini for each subset → weighted Gini → selected split.
K-means	Initial centroids → distances → assignments → new centroids → repeat → final clusters.
Confusion matrix	TP, TN, FP, FN → accuracy, precision, recall/sensitivity and specificity; include F1 if asked.

Weightage note: “Very high / high / medium” means preparation priority based on the uploaded papers, not a prediction of exact marks. The official syllabus and lecturer’s module mapping take precedence, especially for PYQ-linked topics whose module placement is unclear.

Source basis: supplied DSML Module 1–5 notes; Series Test I paper; S3 previous year question papers.pdf, DSML exam pages 1–7. Other course papers in the same PDF were excluded.